

Chapter 2

Numerical Integrators

At its most basic level, molecular dynamics is about mapping out complicated point sets using trajectories of a system of ordinary differential equations (or, in Chaps. 6–8, a stochastic-differential equation system). The sets are typically defined as the collection of probable states for a certain system. In the case of Hamiltonian dynamics, they are directly associated to a region of the energy landscape. The trajectories are the means by which we efficiently explore the energy surface. In this chapter we address the design of numerical methods to calculate trajectories.

When we use the equations of motion for an atomic system and compute a trajectory, we are producing what we hope to be a representative path of the system. For the moment, we discount any external interactions (such as due to heating at a boundary or other driving forces), so we can think of the system as a closed, Hamiltonian dynamics model. This introduces the *microscopic* perspective which concerns the detailed description of the instantaneous atomic positions and velocities as time is varied. These microscopic paths are the cornerstone of statistical mechanical theory, which is the tool that we will eventually develop for understanding molecular systems at a more abstract level, thus it is essential in developing a computational methodology that we have an understanding of how to generate trajectories reliably and efficiently.

The challenge before us is to compute solutions of

$$\dot{q} = M^{-1}p, \quad \dot{p} = F(q) = -\nabla U(q),$$

or, more compactly, with z representing the collection of all positions and momenta,

$$\dot{z} = f(z), \quad f(z) = J\nabla H, \quad (2.1)$$

where $J = \begin{bmatrix} \mathbf{0} & I \\ -I & \mathbf{0} \end{bmatrix}$, and $H = p^T M^{-1} p / 2 + U(q)$ is the Hamiltonian.

$\det \mathcal{G}'_h = 1$. Thus the asymmetric variant of the Euler method is area preserving, even though the standard Euler method is not.

It would be valuable to have a general method for deriving volume conserving methods. It turns out that volume conservation is, itself, most readily obtained as a consequence of a more fundamental property of Hamiltonian flows, the conservation of the symplectic form.

2.3.3 The Symplectic Property

Let $m = 2N_c$. A symplectic map $\Phi : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is one that preserves the symplectic differential 2-form. The simplest way to write this is as the following algebraic condition on the Jacobian matrix of Φ :

$$\Phi'^T J \Phi' = J.$$

Recall that the Jacobian matrix is the $m \times m$ matrix of partial derivatives of the components of Φ , $(\Phi')_{ij} = \frac{\partial \Phi_i}{\partial z_j}$, while

$$J = \begin{bmatrix} \mathbf{0} & I \\ -I & \mathbf{0} \end{bmatrix}.$$

So the equation $\Phi'^T J \Phi' = J$ represents a large number ($4N_c^2$) of equations.

To explain the origin of this formula, we need a brief diversion into the world of differential forms. This language will also be useful in Chap. 4, where we consider constrained systems.

A 1-form α defined on $\mathbb{R}^m$ is a family of linear mappings from $\mathbb{R}^m$ to $\mathbb{R}$, defined for each point of $\mathbb{R}^m$. Let $\mathbf{a} : \mathbb{R}^m \rightarrow \mathbb{R}^m$, then we may define a one-form associated to this vector by $\alpha(\mathbf{x})(\xi) = \mathbf{a}(\mathbf{x})^T \xi$.

The differential of a function $g : \mathbb{R}^m \rightarrow \mathbb{R}$, denoted dg , is a family of linear mappings (one for each point in phase space) from vectors $\xi \in \mathbb{R}^m$ into the reals defined by

$$dg(\mathbf{q}, \mathbf{p})(\xi) = \nabla g(\mathbf{q}, \mathbf{p})^T \xi.$$

So, denoting the i th position coordinate by q_i , we have $dq_i(\xi) = \xi_i$; the differential is thus an example of a 1-form.

On the other hand the so called *wedge product* of 1-forms α , β , is an example of a 2-form, at any point in phase space it can be viewed as a quadratic form, i.e. a scalar valued function of two vectors which is linear in each argument. It is written $\alpha \wedge \beta$ and is defined, for vectors $\xi, \eta \in \mathbb{R}^m$ by

$$\alpha \wedge \beta(\xi, \eta) = \alpha(\xi)\beta(\eta) - \alpha(\eta)\beta(\xi).$$